

ELEC4631 Solutions Tutorial 2

1. (a) Quadratic form:

$$f(x) = x^T \begin{bmatrix} 4 & 3 & 0 \\ 3 & 4 & 4 \\ 0 & 4 & 4 \end{bmatrix} x,$$

with $x = (x_1, x_2, x_3)^T$.

- (b) The characteristic equation of

$$A = \begin{bmatrix} 4 & 3 & 0 \\ 3 & 4 & 4 \\ 0 & 4 & 4 \end{bmatrix}$$

is $c_A(\lambda) = (\lambda - 4)(\lambda + 1)(\lambda - 9)$ and eigenvalues are $\lambda_1 = 4$, $\lambda_2 = -1$, and $\lambda_3 = 9$. Corresponding normalised eigenvectors can be chosen to be $v_1 = (4/5, 0, -3/5)^T$, $v_2 = (3/\sqrt{50}, -5/\sqrt{50}, 4/\sqrt{50})^T$, and $v_3 = (3/\sqrt{50}, 5/\sqrt{50}, 4/\sqrt{50})^T$. The transformation V is

$$V = \begin{bmatrix} 4/5 & 3/\sqrt{50} & 3/\sqrt{50} \\ 0 & -5/\sqrt{50} & 5/\sqrt{50} \\ -3/5 & 4/\sqrt{50} & 4/\sqrt{50} \end{bmatrix}.$$

Then $g(z) = f(Vz) = 4z_1^2 - z_2^2 + 9z_3^2$.

- (c) A has eigenvalues of different signs, therefore it is an indefinite quadratic form.

2. **Solution method I.** $f(x) = x^T A_\epsilon x$, with

$$A_\epsilon = \begin{bmatrix} 1 & \epsilon \\ \epsilon & 1 \end{bmatrix},$$

f is positive definite if and only if A_ϵ is positive definite. $c_{A_\epsilon}(\lambda) = (\lambda - 1 + \epsilon)(\lambda - 1 - \epsilon)$. For all roots of A_ϵ to be positive (we know all roots must be real since A_ϵ is symmetric) requires $-1 < \epsilon < 1$ (but the problem already specifies that $\epsilon > 0$).

$g(x) = x^T B_\epsilon x$, with

$$B_\epsilon = \begin{bmatrix} -\epsilon & -\epsilon/2 \\ -\epsilon/2 & -(1-\epsilon) \end{bmatrix},$$

g is negative definite if and only if B_ϵ is negative definite. $c_{B_\epsilon}(\lambda) = \lambda^2 + \lambda + \epsilon - \frac{5}{4}\epsilon^2$. The roots are $\frac{1}{2}(-1 \pm \sqrt{1 - 4\epsilon + 5\epsilon^2})$. Note that $1 - 4\epsilon + 5\epsilon^2$ cannot be negative (meaning complex roots) since B_ϵ is symmetric. For all roots of B_ϵ to be negative requires $1 - 4\epsilon + 5\epsilon^2 < 1$. Since $\epsilon > 0$ has been specified in problem, this means that the requirement is $0 < \epsilon < 4/5$.

Combining the requirement for both f and g to be positive and negative definite, respectively, gives the condition $0 < \epsilon < 4/5$.

Solution method II. By completion of squares, write

$$f(x) = (x_1 + \epsilon x_2)^2 + (1 - \epsilon^2)x_2^2$$

so f is a positive definite quadratic form if and only if $-1 < \epsilon < 1$.

Similarly, write

$$g(x) = -\epsilon(x_1 + \frac{1}{2}x_2)^2 - (1 - \frac{5}{4}\epsilon)x_2^2$$

so g is a negative definite quadratic form if and only if $\epsilon > 0$ (already specified in the problem) and $1 - \frac{5}{4}\epsilon > 0$. That is, $0 < \epsilon < 4/5$.

So, for f and g to be simultaneously positive and negative definite, respectively, we require $0 < \epsilon < \min(1, 4/5) = 4/5$.

3. (a) Eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 3$. Diagonalising matrix is

$$V = \begin{bmatrix} 1 & 8 \\ 1 & 12 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} e^A &= V \text{diag}(e^{-1}, e^3) V^{-1}, \\ &= \frac{1}{4} \begin{bmatrix} 12e^{-1} - 8e^3 & -8e^{-1} + 8e^3 \\ 12e^{-1} - 12e^3 & -8e^{-1} + 12e^3 \end{bmatrix}. \end{aligned}$$

- (b) Eigenvalues are $\lambda_1 = 1 + j$ and $\lambda_2 = 1 - j$. Diagonalising matrix is

$$V = \begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix}.$$

Using the identities $\cos x = (e^{jx} + e^{-jx})/2$ and $\sin x = (e^{jx} - e^{-jx})/(2j)$,

$$e^A = V \begin{bmatrix} e^{1+j} & 0 \\ 0 & e^{1-j} \end{bmatrix} V^{-1} = e \begin{bmatrix} \cos 1 & \sin 1 \\ -\sin 1 & \cos 1 \end{bmatrix}.$$

- (c) $c_A(\lambda) = \lambda^3 + 0.5\lambda^2 - 0.5\lambda = \lambda(\lambda^2 + 0.5\lambda - 0.5)$. Eigenvalues are $\lambda_1 = 0$, $\lambda_2 = -1$ and $\lambda_3 = 0.5$. Corresponding eigenvectors can be chosen to be $v_1 = (1, 0, 0)^T$, $v_2 = (1, -1, 1)^T$, and $v_3 = (4, 2, 1)^T$, so

$$V = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 4 \\ 0 & -1 & 2 \\ 0 & 1 & 1 \end{bmatrix}.$$

Using expansion by minors based on the third row of V , the determinant of V is $\det(V) = -1 \times (1 \times 2) + 1 \times (1 \times -1) = -2 - 1 = -3$. The transpose of the adjugate matrix of V is

$$\text{adj}(V)^T = \begin{bmatrix} -1-2 & 0 & 0 \\ -(1-4) & 1 \times (1-0) & -1 \times (1-0) \\ (1 \times 2 + 1 \times 4) & -1 \times (1 \times 2 - 0) & 1 \times -1 - 0 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 0 \\ 3 & 1 & -1 \\ 6 & -2 & -1 \end{bmatrix}.$$

So

$$V^{-1} = \frac{1}{\det(V)} \text{adj}(V) = -\frac{1}{3} \begin{bmatrix} -3 & 3 & 6 \\ 0 & 1 & -2 \\ 0 & -1 & -1 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} e^A &= V \text{diag}(1, e^{-1}, e^{0.5}) V^{-1} \\ &= -\frac{1}{3} \begin{bmatrix} -3 & 3 + e^{-1} - 4e^{0.5} & 6 - 2e^{-1} - 4e^{0.5} \\ 0 & -e^{-1} - 2e^{0.5} & 2e^{-1} - 2e^{0.5} \\ 0 & e^{-1} - e^{0.5} & -2e^{-1} - e^{0.5} \end{bmatrix} \end{aligned}$$

4. $H(s)$ can be decomposed as:

$$H(s) = \frac{3s^2 + 21s + 33}{s^3 + 9s^2 + 23s + 15} + 1$$

- (a) Control canonical form is

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -15 & -23 & -9 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 33 & 21 & 3 \end{bmatrix} x(t) + u(t). \end{aligned}$$

See block diagram on page 6.

(b) By partial fraction expansion, we have

$$H(s) = \frac{15/8}{s+1} + \frac{3/4}{s+3} + \frac{3/8}{s+5} + 1,$$

modal canonical form is:

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -5 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 15/8 & 3/4 & 3/8 \end{bmatrix} x(t) + u(t).\end{aligned}$$

See block diagram on page 6.

(c) Observer canonical form is

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 0 & -15 \\ 1 & 0 & -23 \\ 0 & 1 & -9 \end{bmatrix} x(t) + \begin{bmatrix} 33 \\ 21 \\ 3 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x(t) + u(t).\end{aligned}$$

See block diagram on page 6.

5. (a)

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} -1 & 1 \end{bmatrix} x(t).\end{aligned}$$

(b) The A matrix has eigenvalues -3 and 1, and has the decomposition

$$A = \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix} \text{diag}(-3, 1) \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix}^{-1},$$

so

$$e^{At} = \frac{1}{4} \begin{bmatrix} e^{-3t} + 3e^t & -e^{-3t} + e^t \\ -3e^{-3t} + 3e^t & 3e^{-3t} + e^t \end{bmatrix}.$$

The solution to state space equation is

$$\begin{aligned}x(t) &= \frac{1}{4} \begin{bmatrix} e^{-3t} + 3e^t & -e^{-3t} + e^t \\ -3e^{-3t} + 3e^t & 3e^{-3t} + e^t \end{bmatrix} x_0 + \frac{1}{4} \int_0^t \begin{bmatrix} -e^{-3(t-s)} + e^{t-s} \\ 3e^{-3(t-s)} + e^{t-s} \end{bmatrix} u(s) ds, \\ y(t) &= \begin{bmatrix} -e^{-3t} & e^{-3t} \end{bmatrix} x_0 + \int_0^t e^{-3(t-s)} u(s) ds.\end{aligned}$$

(Note that $y^{(1)}(t) = \begin{bmatrix} 3e^{-3t} & -3e^{-3t} \end{bmatrix} x_0 - 3 \int_0^t e^{-3(t-s)} u(s) ds + u(t)$.)

(c)

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 3 \\ 1 & -2 \end{bmatrix} x(t) + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t).\end{aligned}$$

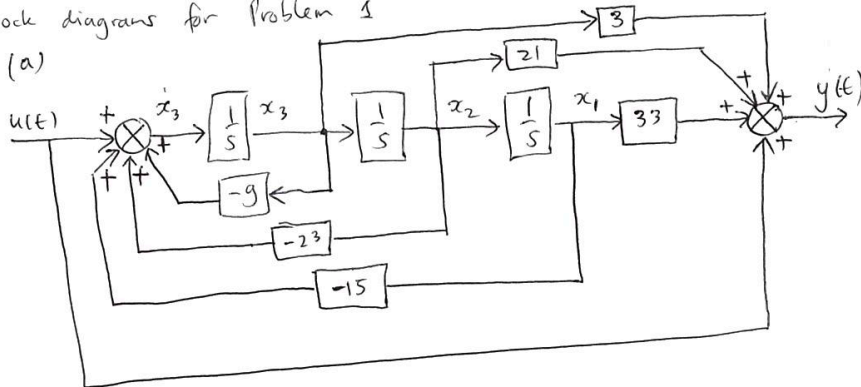
(d) The solution to the state space equation is

$$\begin{aligned}x(t) &= \frac{1}{4} \begin{bmatrix} e^{-3t} + 3e^t & -3e^{-3t} + 3e^t \\ -e^{-3t} + e^t & 3e^{-3t} + e^t \end{bmatrix} x_0 + \int_0^t \begin{bmatrix} -e^{-3(t-s)} \\ e^{-3(t-s)} \end{bmatrix} u(s) ds, \\ y(t) &= \frac{1}{4} \begin{bmatrix} -e^{-3t} + e^t & 3e^{-3t} + e^t \end{bmatrix} x_0 + \int_0^t e^{-3(t-s)} u(s) ds(t).\end{aligned}$$

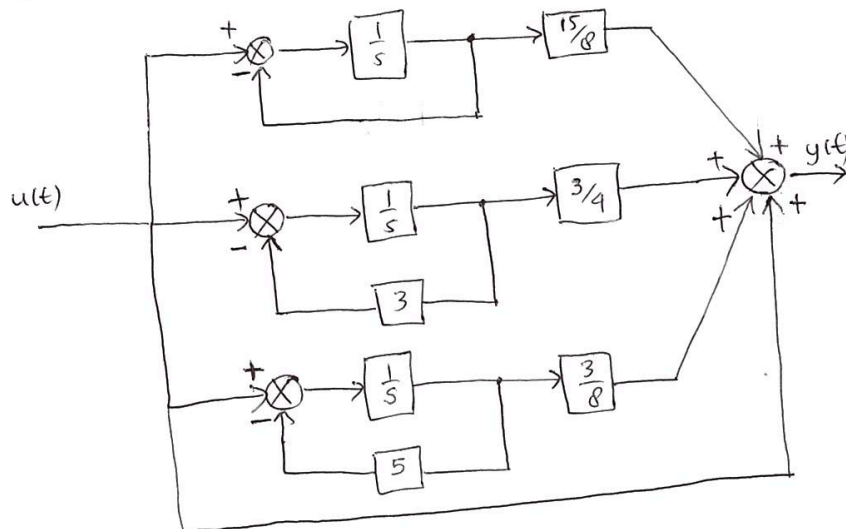
(Note that $y^{(1)}(t) = \frac{1}{4} \begin{bmatrix} 3e^{-3t} + e^t & -9e^{-3t} + e^t \end{bmatrix} x_0 - 3 \int_0^t e^{-3(t-s)} u(s) ds + u(t)$.)

Block diagrams for Problem 1

(a)



(b)



(c)

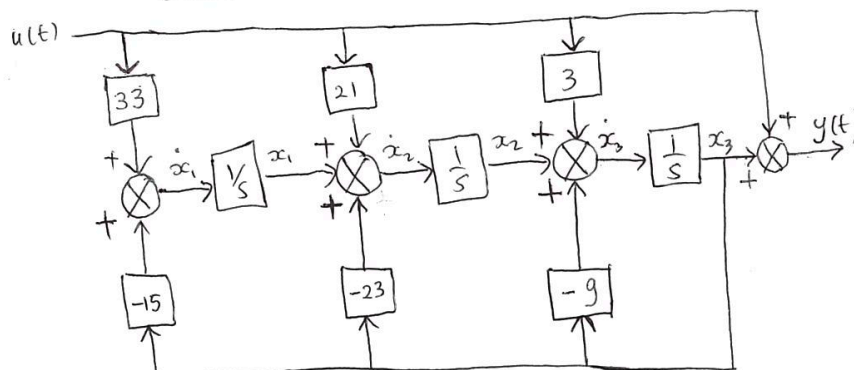


Figure 1: Sketch for Q4.